

Existential Consistency: Measuring and Understanding Consistency at Facebook

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SOSP 2015

- Haonan Lu
- Edit Profile
- FAVORITES
 - Welcome
 - News Feed
 - Messages
 - Events
- APPS
 - Games
 - Find Friends
 - Photos
 - Suggest Edits
 - Pokes
 - Saved
 - Translate Facebook
- EVENTS
 - First Commenter Wins...
 - Create Event
- PAGES
 - Pages Feed
 - Like Pages
 - Create Page
- GROUPS
 - Create Group

Update Status | Add Photos/Video | Create Photo Album

What's on your mind?

Friends | Post

Viewing most recent stories · Back to top stories

30 First Commenter Wins... on October 6

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**Haonan Lu** is going to an event.
2 hrs · 

06 OCT

First Commenter Wins *Free* Oculus!
Tue 12 AM · Monterey, California, USA
Symposium on Operating Systems Principles is going

Going ↕

 Like  Comment  Share





Like Comment Share



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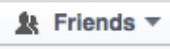
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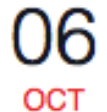
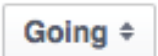
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GAMES YOU MAY LIKE [See All](#)

 **Clash of Kings**
1 million players


 **War Commander**
100,000 players


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 **Haonan Lu** Mine! yeah~ lucky!



1 min · Like

 Write a comment...  



Haonan Lu

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What's on your mind?

Consistency

Performance

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1 hr ·

06
OCT**First Commenter Wins *Free* Oculus!**

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Going

Like

Comment

Share

**Wyatt Lloyd** I wouldn't mind taking one.

1 min · Like

**Haonan Lu** Mine! yeah~ lucky!

Just now · Like



Write a comment...



10 million players

Play Now

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Fundamental Tension

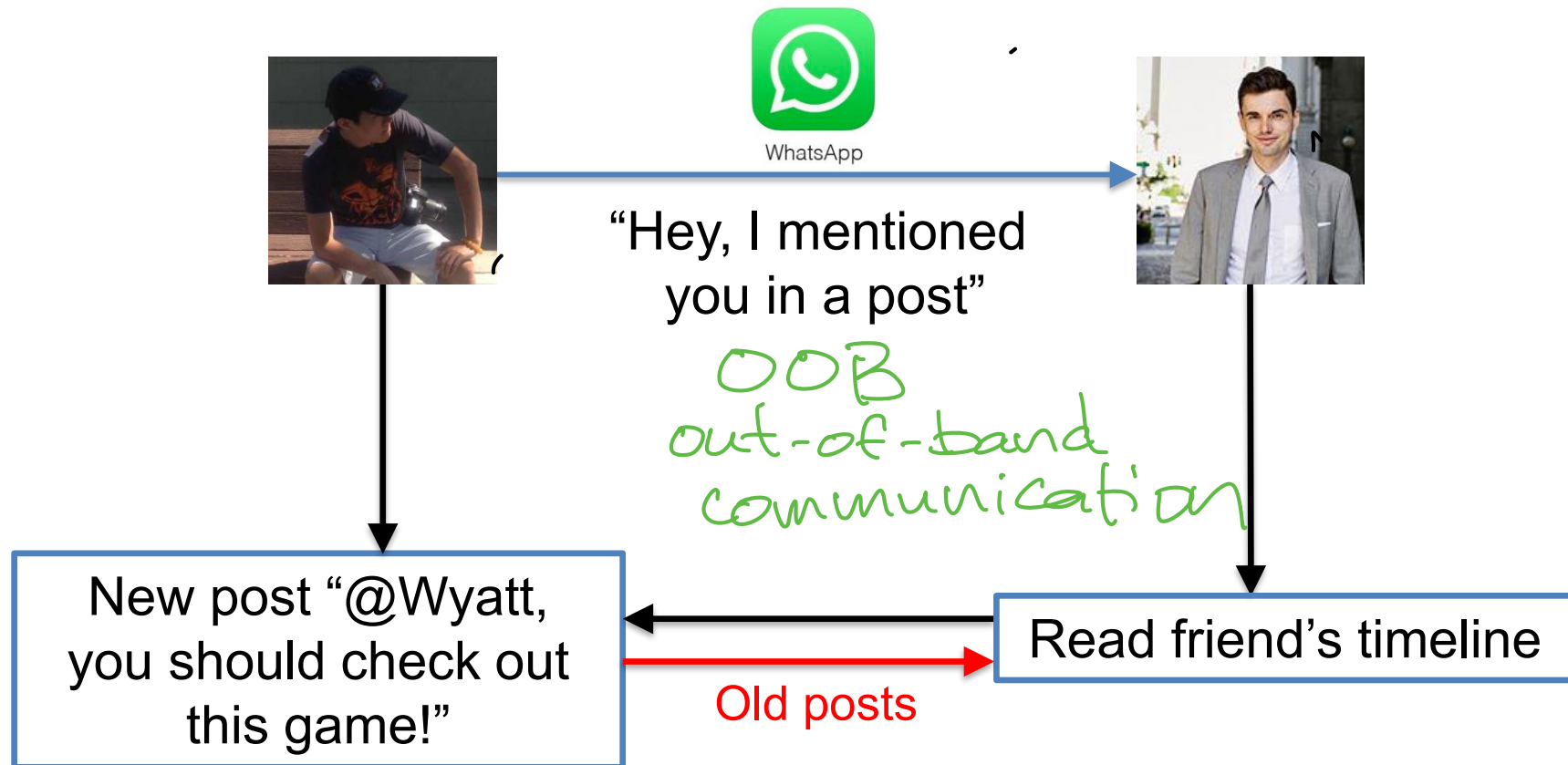
Consistency Performance

- Eliminates anomalies
(Oculus example)
- Makes systems easier to program
- Difficult to quantify
- Lower latency
- Higher throughput
- Simple to quantify

First study of consistency in a large-scale, production system – Facebook TAO

Anomaly: Unexpected Behavior

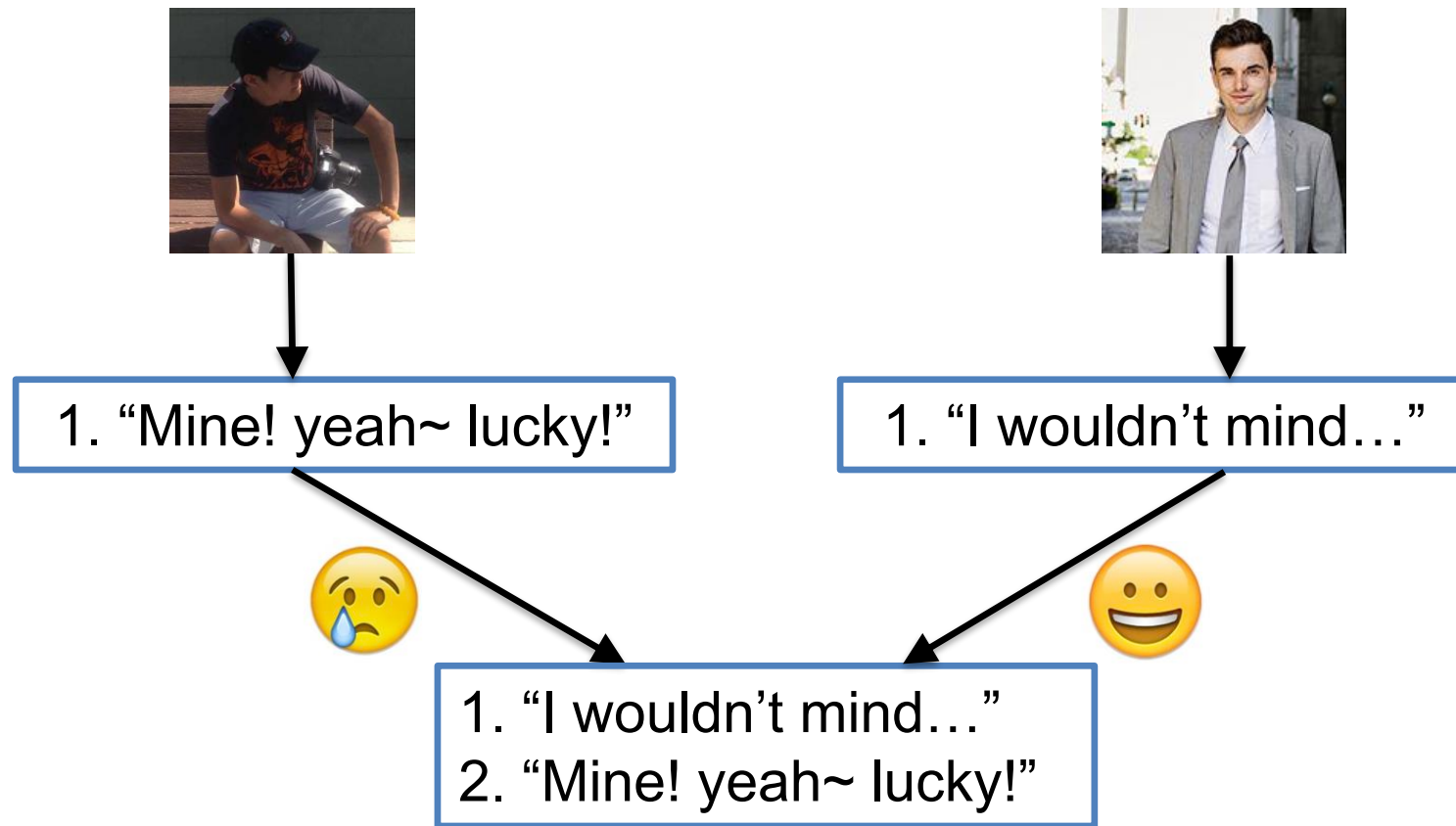
Post Example



1) Doesn't match real time

Anomaly: Unexpected Behavior

Oculus Example



2) Ordering Anomaly

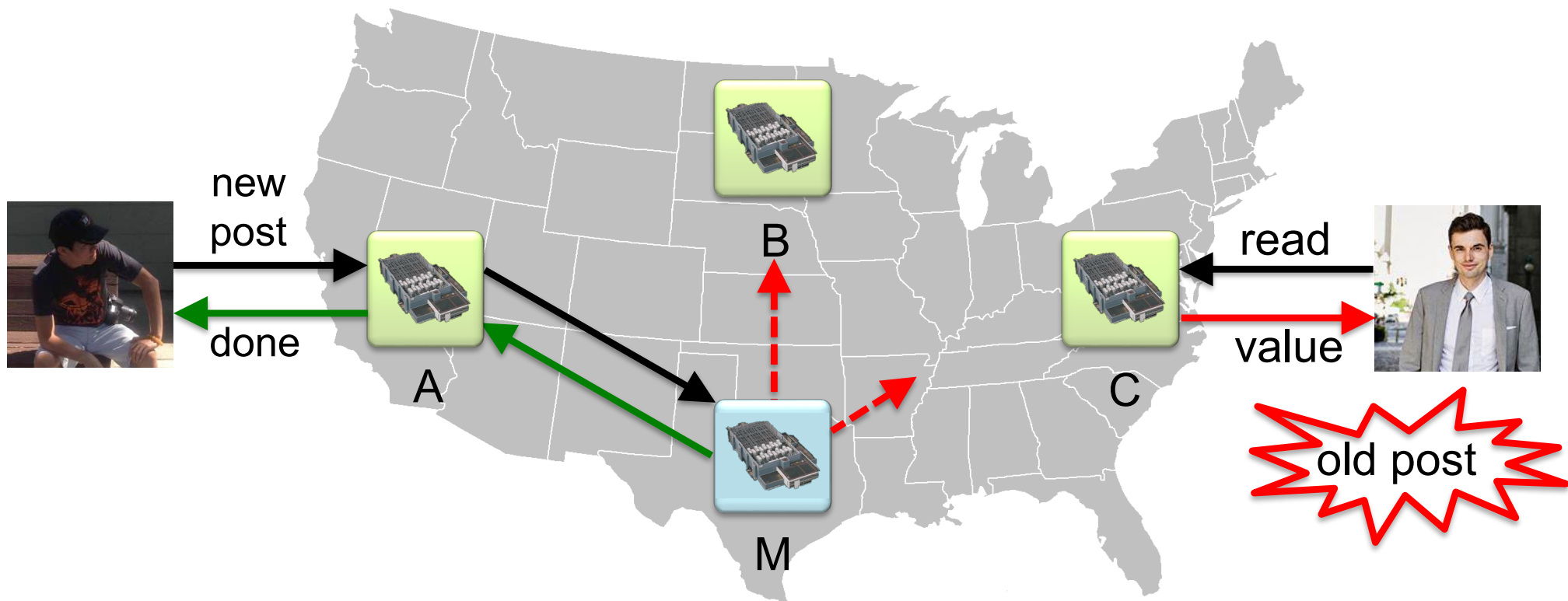
Does Facebook have
consistency anomalies?

How many?

What type?

TAO: Eventually Consistent Cache

Vulnerability window: time during asynchronous replication when anomalies can happen



Quantifying Anomalies

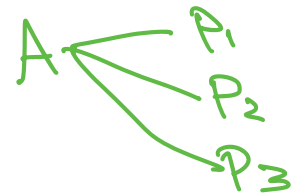
- How often do anomalies occur?
 - Collect trace of requests to TAO
- What consistency would prevent them?
 - Run anomaly checkers on the trace

Trace Collection

- Collect trace on web servers
- Challenges in tracing production system
 - 1) – Volume of requests
 - 2) – Time skew between web servers
 - ~~– Missing requests~~ →
 - sanity check w/ another logging system
 - ignore reads if no corresponding write

Challenge: Volume of Requests

- Billions of requests per second [ATC '13]
 - Too many to log



- Sample on objects
 - Object: vertex in social graph
 - Log all requests to objects in sample
 - Sufficient for local consistency models
- (users, posts)*
and edges: who created post

Local Property Enables Sampling

- “... the system as a whole satisfies P whenever each individual object satisfies P .”^[1]

Local consistency models can be checked on a per object basis

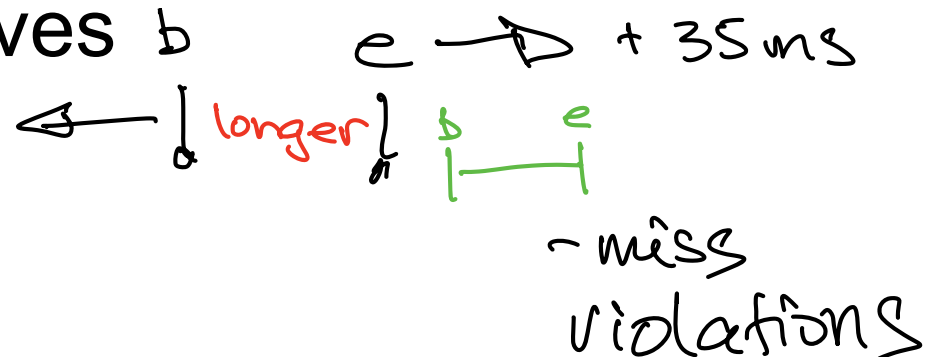
- Local
 - Linearizability
 - Per-Object Sequential
 - Read-After-Write
- ~~Non-local~~
 - ~~Strict Serializability~~
 - ~~Causal~~

[1] M. P. Herlihy and J. M. Wing “Linearizability: A Correctness Condition for Concurrent Objects.” ACM TOPLAS, 1990

Challenge: Time Skew

- Time skew across web servers
 - 99.9 percentile for 1 week: 35ms
- Add time skew to request's duration
 - More overlapped requests
 - Eliminates false positives

- wait identify as
"violations" events that
weren't violations



- underestimate violations

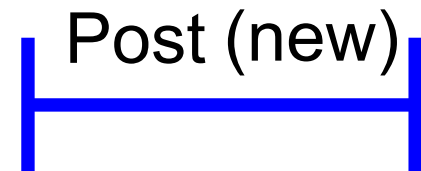
Logging Details

- Logged information:

- Start time
 - Finish time
- } Determine real time ordering of requests

- Read or write

- Value: match read with write



- Sampling rate: 1 out of 1 million objects
~ 100% of requests to sampled objects

Trace Statistics

- 12 days (8/20 – 8/31)
- 17 million objects
- 3 billion requests

Check Trace for Anomalies

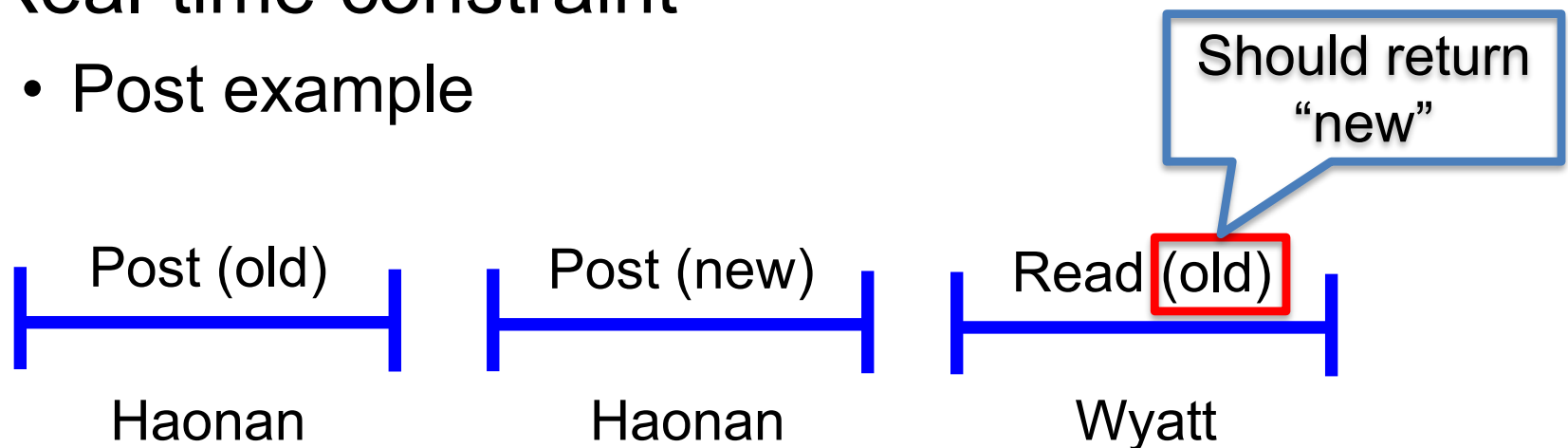
- Linearizability checker
 - Paxos provides
- Per-Object Sequential checker
 - PNUTS provides
- Read-After-Write checker
 - TAO provides within a cluster

Linearizability

- Strongest non-transactional consistency

- Real-time constraint

- Post example



- Total order constraint

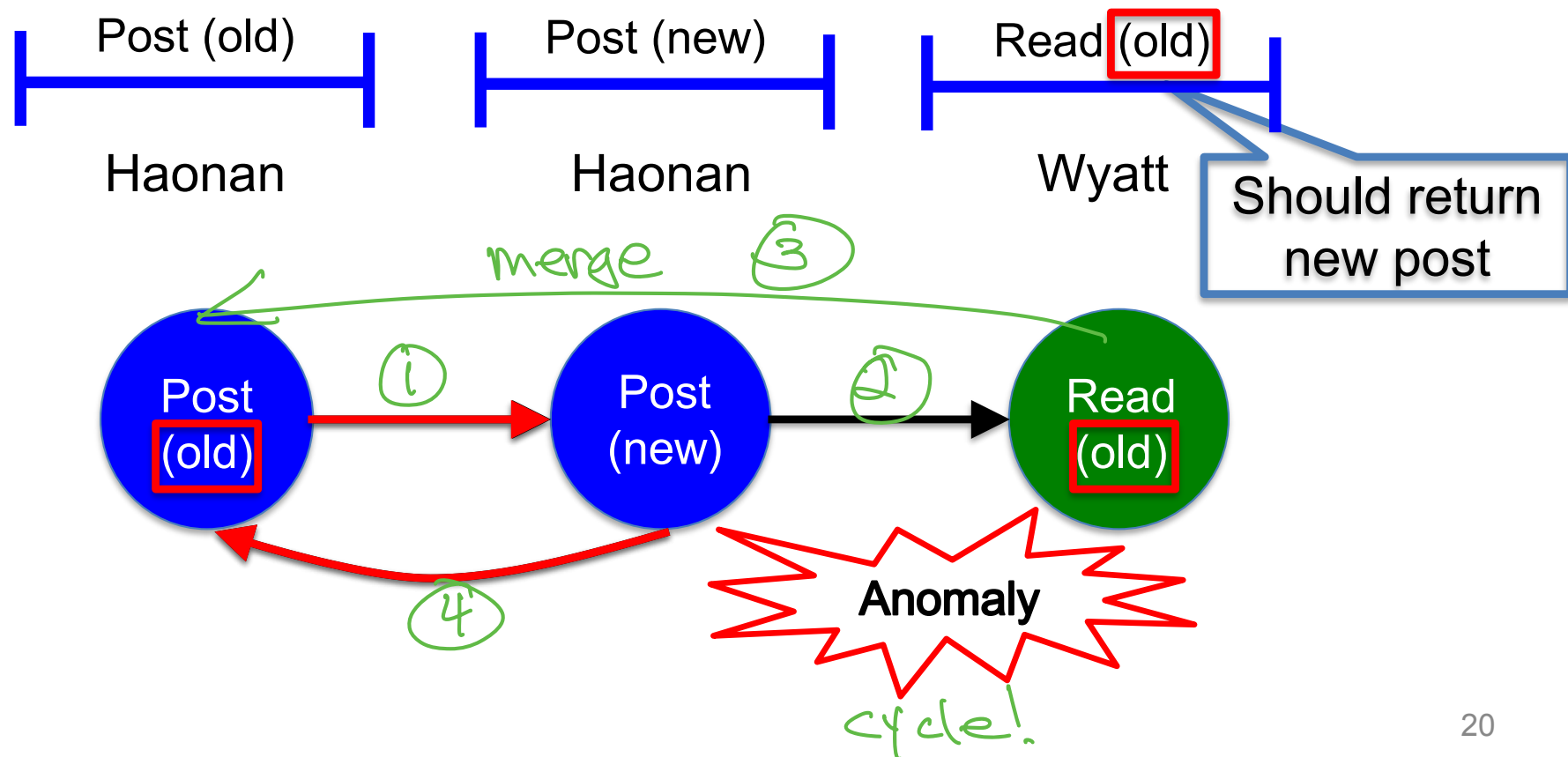
- Oculus example!

Linearizability Checker

- Graph captures state transitions
 - Vertex: write operations
 - Edge: real-time order
- Merge read with its write
 - Captures state transitions seen by users
- Anomaly if merge causes a cycle
 - Cycle indicates user's view $\neq$ system view

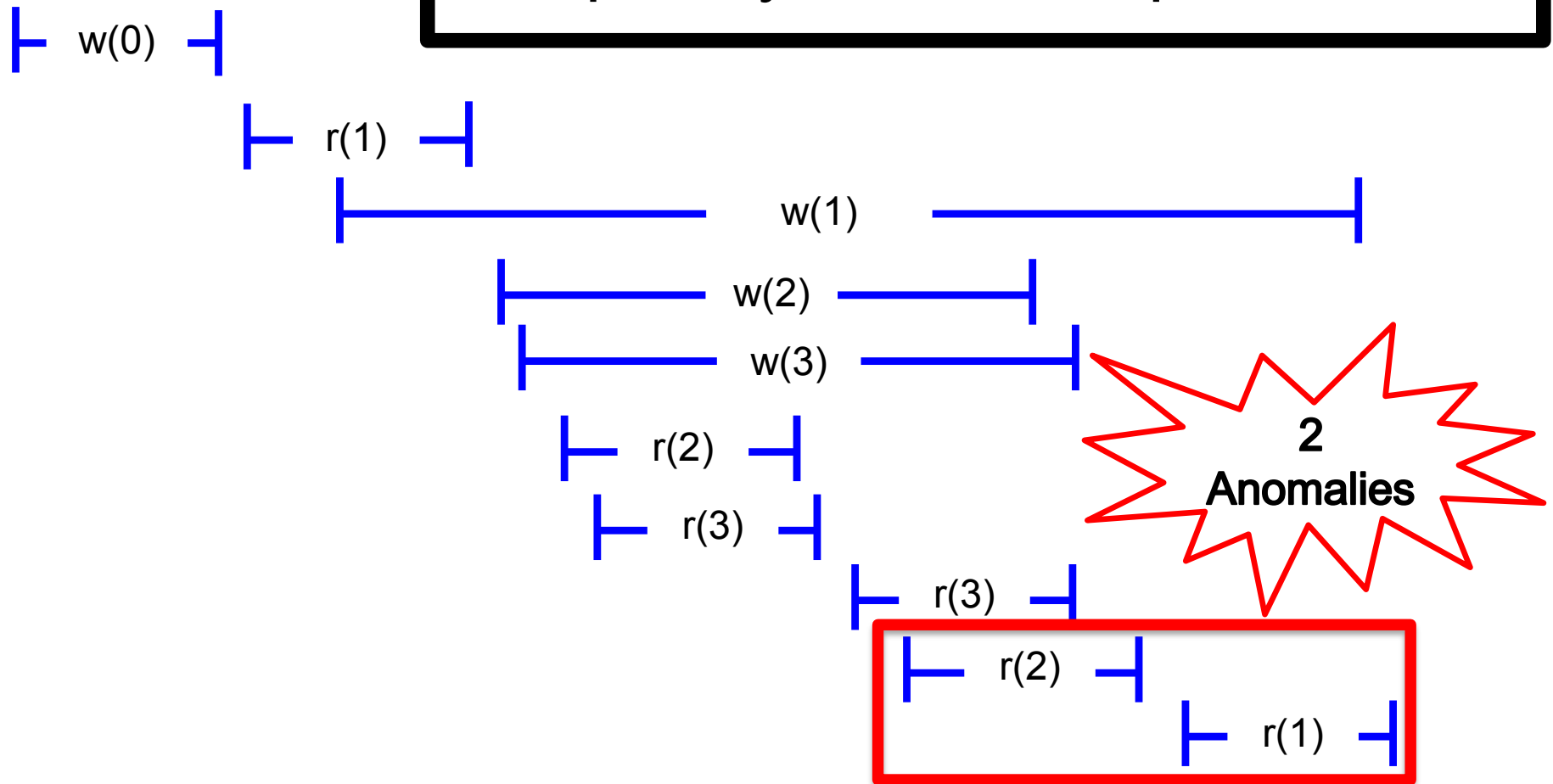
Linearizability Checker

- Captures real-time constraint
 - Read should return new post instead



More Complex Cases

<http://tinyurl.com/sosp15-demo>



Result Overview

- Linearizability
- Per-Object Sequential
- Read-After-Write
- Bounds on non-local consistency models

Anomalies found for all consistency models
– adopting them would have benefits

Linearizability Results

- 5 anomalies per million reads
 - Prevented by Paxos-based implementation
- Upper bound on TAO anomalies
 - Strongest consistency we checked

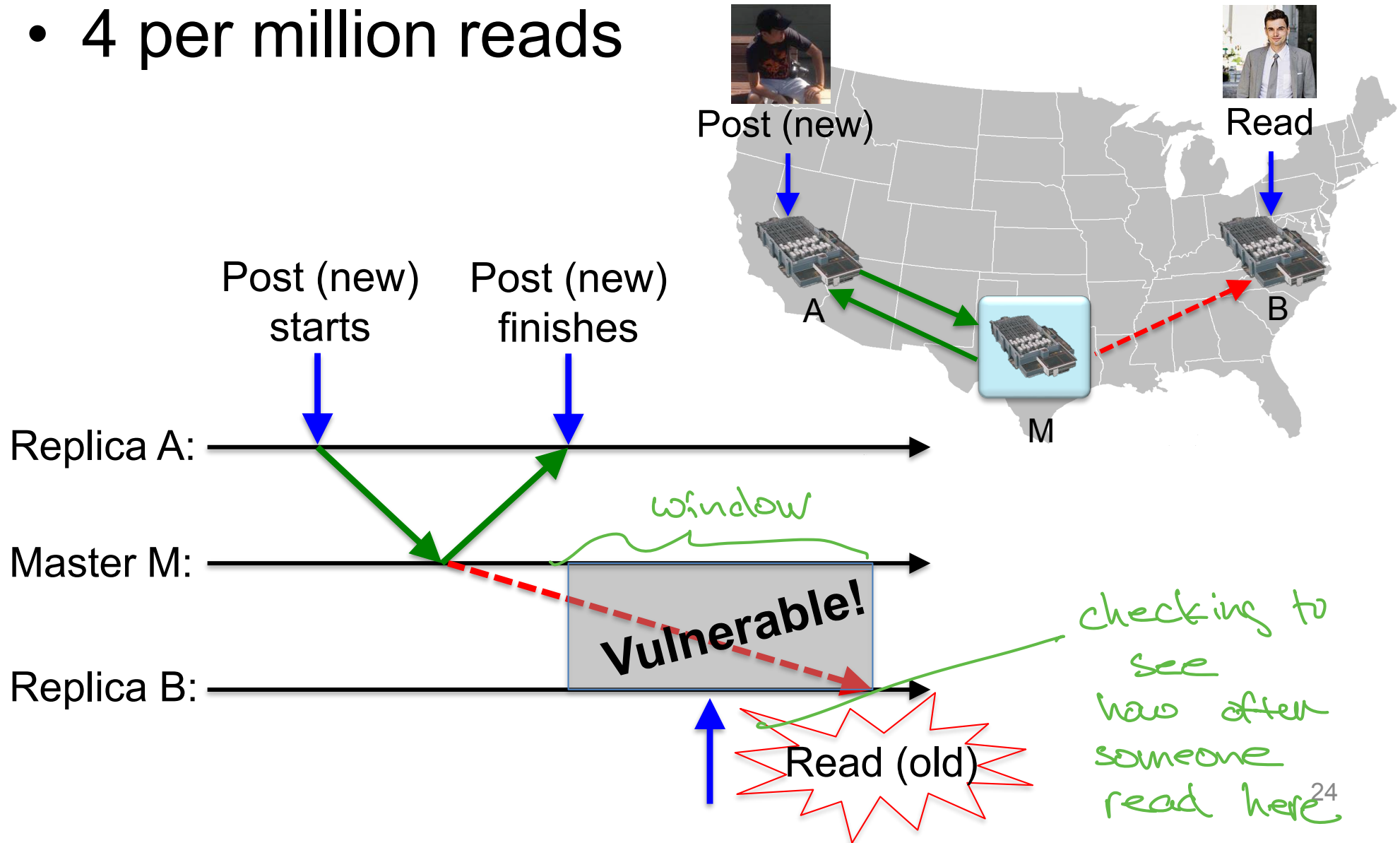
TAO is highly consistent

Of those
5 million:

Linearizability Results

Real-Time Constraint Violations

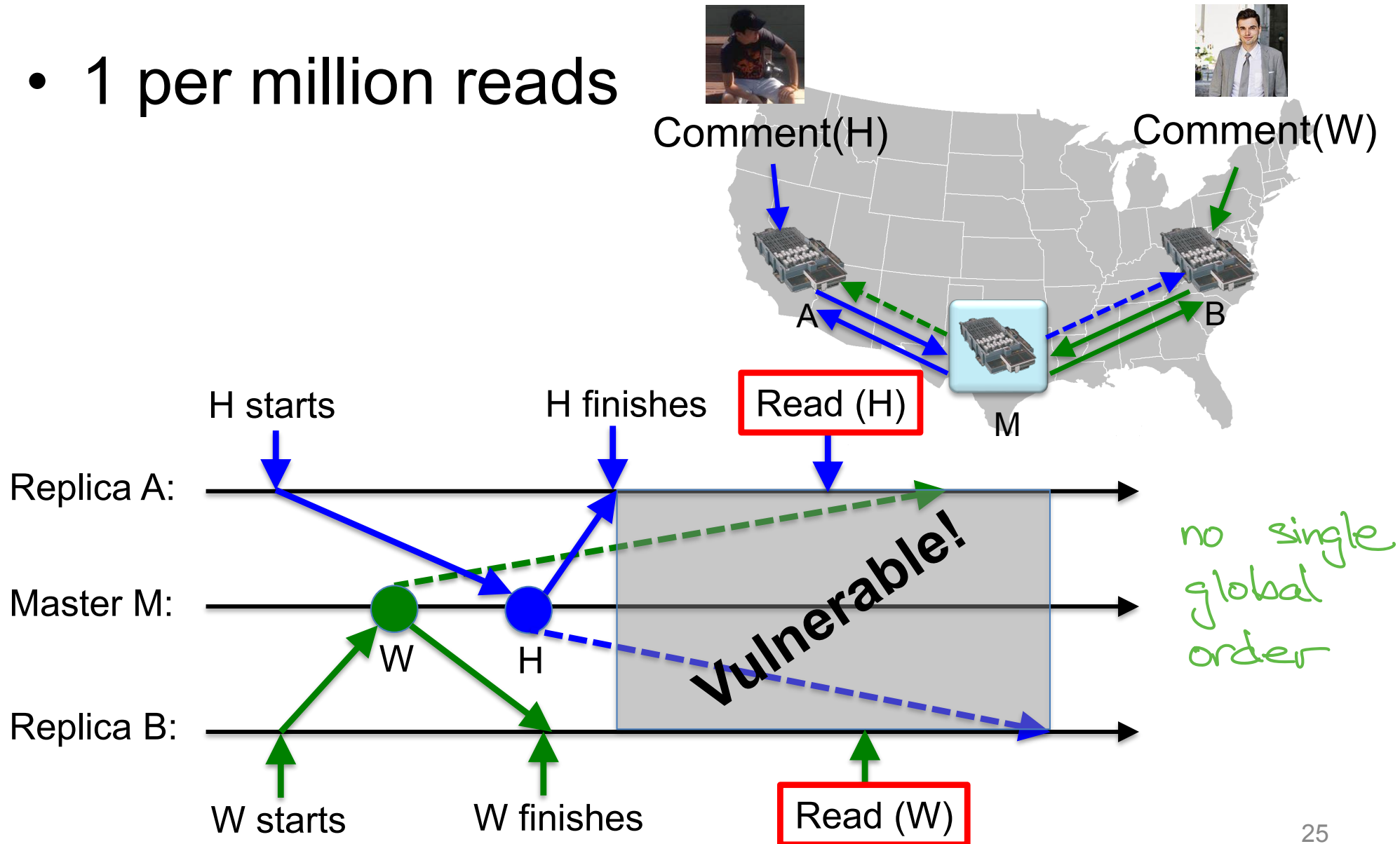
- 4 per million reads



of 5/million Linearizability Results

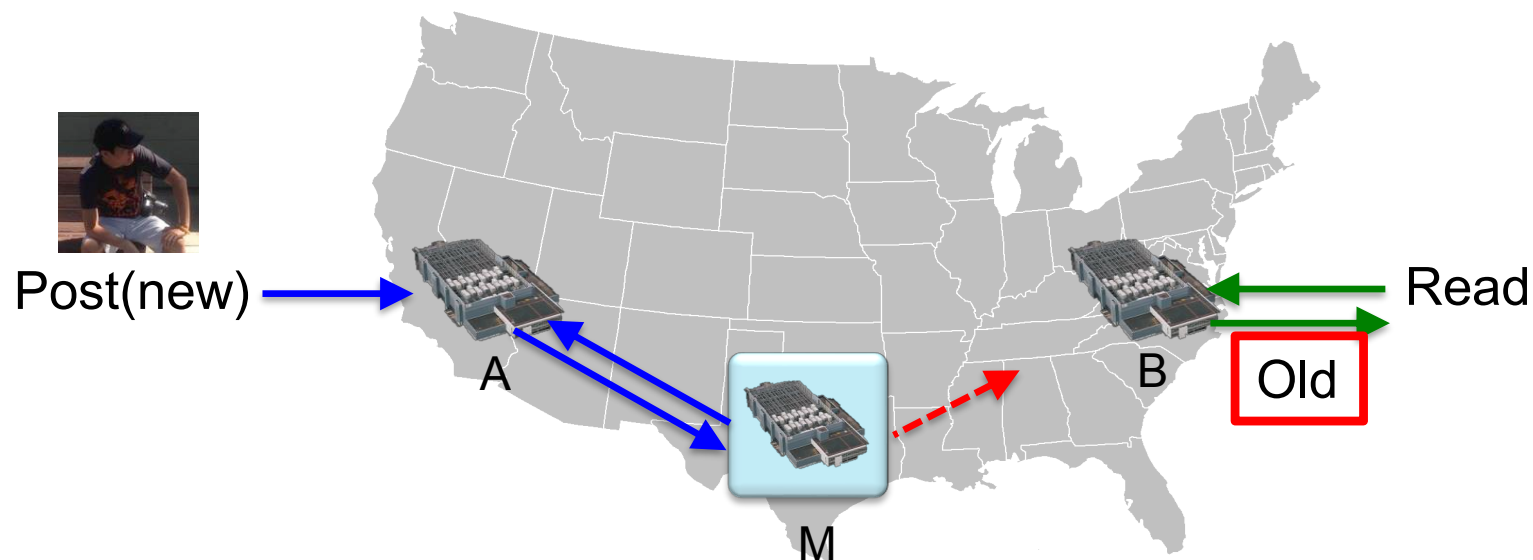
Total Order Constraint Violations

- 1 per million reads



Per-Object Sequential Results

- 1 anomaly per million reads
 - Total order constraint – *not real time*
 - User session constraint (1 per 10 million)
 - Users should see their writes (*same sequential stream; reads after writes*)



get redirected to different replica

Real-Time Consistency Monitor

- Checkers cannot run in real-time
- Φ -consistency
 - Measure convergence of replicas
- A real-time health monitor
 - Alarms when a replica falls behind

Conclusion

- Benefits of consistency are hard to quantify
 - First study of a large-scale production system
- Measure Facebook's TAO system
 - Collect trace and run anomaly checkers
 - Real-world challenges
- Results
 - TAO is highly consistent
 - Benefits of adopting stronger consistency exist
 - Research should provide transactions

Interesting Points in Paper

Analysis:

Only objects that have both read + write requests can exhibit anomalies

Vertices Objects Requests	Both R+W 10.7% 25.7%
Edges Objects Requests	4.1% 22.8%

Small # of edge types account for most anomalies

Fewer anomalies than they expected

- Sensitive apps avoid TAO
- Sensitive apps can mark reads as critical $\rightarrow$ forces requests through master $\rightarrow$ linearizable

Interesting Developer Errors

- Look-aside cache - Explicitly managed
 - Developers manually write data to cache + issue invalidations

- 1) Cache failure or error code
- 2) Forget to invalidate negative caching
- 3) Cache time-dependant results
- 4) Cache cached results - invalidations don't occur

Audience Questions - Watch recorded talk

1) What type of consistency should be provided?

- TAO is highly consistent
- But, strong consistency is more
- No recommendation!

2) Time scale skew gives false negatives?

- Eliminates false positives!
- Yes, false negatives
- Look @ other adjustments

3) Algorithms exist for checking consistency

- We cited + do not claim novelty
- Contribution is measurement results + system challenges

4) Consistency should be relaxed for extreme situations - shutting down data center; why not study during fail over?

- That would be interesting!

5) 5 anomalies of 1 million; is that ok for Facebook? How many user interactions are affected?

- Should only do stronger if low perf overhead
- Don't know impact on end users